ETHEL OGALLO

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LinkedIn | Github | ePortfolio

SUMMARY

GeoAI and Earth Observation research interested in the intersection of AI in Earth Observation (AI4EO), with a focus on environmental monitoring . Experienced in automated workflows for processing and analyzing Earth observation data using Python and R.

EDUCATION

Copernicus Master in Digital Earth **Oct 2024 - Sept 2026**
University of Salzburg, Austria & University of South Brittany , France
Specialization: GeoData Science
Thesis: Towards Super-resolution of Thermal Infrared Imagery to assess temperature spatial heterogeneity in rivers.

Bachelor of Applied Science in Geoinformatics
The Technical University of Kenya, Kenya

PROFESSIONAL EXPERIENCE

Remote Sensing Intern **Jul 2024 -Aug 2024**
The Center for Remote Sensing of Land Surfaces (ZFL); Bonn, Germany

- Developed machine learning workflow for wetland delineation and classification of Lake Baringo wetland in Kenya under the WetlandHealth4UNgoals project. [GitHub Repository]

Data Science Research Assistant **Feb 2024 - Sept 2024**
CEMA Africa; Nairobi, Kenya

- Modeled healthcare accessibility by estimating catchment areas and travel time to facilities, using geostatistical techniques in R, communicating the accessibility visually via an interactive dashboard to support health planning.

Data Science Intern **Aug 2023 - Jan 2024**
CEMA Africa; Nairobi, Kenya

- Analyzed spatio-temporal patterns of cholera outbreak focusing on socio-demographic and environmental factors.
- Mapped cholera hotspots at the ward level, offering actionable insights for targeted interventions and improved disease control strategies.

Environmental Engineer **Nov 2020 - Dec 2022**
Centunion East Africa; Nairobi, Kenya

- Developed, implemented, and maintained a robust environmental management system (EMS), conducting 4 internal audits to ensure compliance with ISO standards and local regulations.

GIS Research Assistant **Mar 2020 - Aug 2020**
Oakar Services; Nairobi, Kenya

- Collected and managed spatial and non-spatial data for a farmer registration project in Kiambu County, using mobile client configuration tools.

SCIENTIFIC ACTIVITIES & PROJECTS

- **ICPR (Feb 2026):** Participated in the ICPR 2026 competition on Kaggle, Beyond Visible Spectrum: AI for Agriculture; awarded 2nd place on the second task "Boosting automatic crop diseases classification using Sentinel satellite data and self-supervised learning"
- **ASPRS 2025 Annual Conference | Washington D.C. (Virtual) (Oct 2025):** Presented the pysarflow tool, demonstrating Python-based automation for SAR data workflows to an international audience.
- **AI4EO Symposium | Rennes, France (11–12 Sept 2025):** Participated in technical sessions on the integration of Artificial Intelligence and Earth Observation
- **MLESM Hackathon | Bonn, Germany (Onsite) (27–29 Aug 2025):** Fine-tuned machine learning models for Earth system modeling; awarded 1st place for innovative geospatial solutions.
- **PySARFlow** (**2024**): An open source python library for preprocessing Sentinel-1 SAR.
- **Next City Summer School | Benicàssim, Spain (Onsite) (June 2024):** Attended a program on building smart and sustainable cities through hands-on exploration of Digital Twin technologies.
- **Quantifying Vulnerability to Natural Hazards: BIP | University of Bucharest, Romania (Hybrid) (Mar -Apr 2024):** Completed a Blended Intensive Programme focused on using multi-criteria decision analysis to quantify socio-environmental vulnerability.
- **Digital Earth Africa (2024):** Contributed Python workflows for analyzing wetland dynamics using large-scale EO datasets and the Open Data Cube.

SKILLS

Programming: Python , R , SQL.
Geospatial: ArcGIS Pro, QGIS, Geoserver, Google Earth Engine, SNAP.
Cloud & DevOps: AWS, Git, GitHub.
Languages: English , Swahili (Native).